

Worksheet 2: The Qubit and Quantum Mechanics Fundamentals

22 September 2026

TA: Haripriya | haripriya.pettugani@unibas.ch

Name: _____

Please write your name on your copy before handing it in.

Questions marked $\star$ are **facultative**. Attempting them counts towards a good effort, but you will not be penalised for leaving them out. Exercise 7 is harder than the rest; go as far as you can with it.

Notation. $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ and $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. For a measurement in the basis $\{|u\rangle, |u^\perp\rangle\}$ the outcome u has probability $|\langle u|\psi\rangle|^2$.

Part A — States and measurement

1. Reconstructing a state from data. A source emits many copies of the same unknown qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Two measurements are performed on separate copies, and the statistics come back as

$$p(0) = \frac{3}{4} \quad \text{under } Z, \quad p(+) = \frac{1}{2} \quad \text{under } X.$$

- Use the Z data to fix $|\alpha|$ and $|\beta|$. Explain why this leaves the state undetermined, and count how many real parameters are still free once the global phase is discarded.
- Take α real and positive and write $\beta = |\beta|e^{i\varphi}$. Impose the X datum and solve for φ . You should find exactly two solutions.
- Predict the Y statistics for each of the two candidates, and state what a third experiment would have to measure to settle which state the source is emitting.

2. An interferometer, worked through. A photon enters a Mach–Zehnder interferometer. The qubit is which arm the photon is in; the beam splitters act as H , and a glass plate in one arm multiplies that arm’s amplitude by $e^{i\varphi}$. The circuit is therefore H , then the phase, then H , applied to $|0\rangle$.

- Track the state through all three stages and show that the detector probabilities are

$$p(0) = \cos^2(\varphi/2), \quad p(1) = \sin^2(\varphi/2),$$

or the two interchanged, depending on which arm you take the plate to be in. Say which arm you chose.

- Using your convention, give the three smallest non-negative values of φ for which the photon arrives at one detector with certainty, splits evenly, and arrives at the other detector with certainty.
- A detector is now placed in one arm, recording which way the photon went. Recompute the two probabilities. State which feature of your calculation in (a) is lost, and why the answer no longer depends on φ .

3. Stern–Gerlach at an arbitrary angle. The sequential experiment is run with the middle magnet at an angle θ to z rather than at 90° . The first magnet is along z and the $|1\rangle$ beam is blocked; the middle magnet is along $\hat{n}(\theta)$ and only its $+$ output is kept; the third magnet is along z again. For a spin, the state selected by the middle magnet is $|n_+\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$.

- (a) What fraction of the atoms leaving the first magnet survives the middle one?
- (b) Of those survivors, what fraction leaves the third magnet through the $|1\rangle$ output?
- (c) Combine the two and show that the fraction of the original beam reaching the $|1\rangle$ output is $\frac{1}{4} \sin^2 \theta$. Find the angle that maximises it and the value there.
- (d) Evaluate your expression at $\theta = 0$ and $\theta = 180^\circ$, and say in each case why the answer is what a classical account would also have predicted.

Part B — Driving a qubit

4. Calibrating a drive from measured data. A new device is characterised by preparing $|0\rangle$, applying a pulse of duration t , and measuring in the computational basis. *On resonance* means that the drive frequency exactly matches the qubit frequency, so the detuning is $\Delta = 0$; in that case the excitation follows

$$p(1) = \sin^2(\Omega t/2).$$

- (a) The measured $p(1)$ first reaches its maximum at $t = 25$ ns. Determine $\Omega/2\pi$ in MHz, and give the duration of the pulse that takes $|0\rangle$ to the equator.
- (b) A second qubit on the same chip is driven with the same pulse amplitude, so it has the same Ω , but its oscillation never rises above $p(1) = 0.64$. A drive detuned by Δ reaches at most $\Omega^2/(\Omega^2 + \Delta^2)$. Extract the ratio Δ/Ω for this qubit.
- (c) A detuned drive rotates the Bloch vector about an axis tilted out of the equatorial plane by $\arctan(\Delta/\Omega)$, at a rate $\sqrt{\Omega^2 + \Delta^2}$ rather than Ω (both are given in the lecture notes, Block 2). Predict the observed oscillation frequency for the second qubit, and the tilt of its rotation axis.
- (d) Explain how the experimenter could have obtained the same value of Δ without ever fitting the contrast, by looking only at the oscillation frequency.

Part C — Limits

5. Copying. A proposed gate C acts on an unknown qubit together with a blank one, and is known to satisfy

$$C|0\rangle|0\rangle = |0\rangle|0\rangle, \quad C|1\rangle|0\rangle = |1\rangle|1\rangle.$$

- (a) Apply C to $|+\rangle|0\rangle$ using only linearity, and write down the resulting two-qubit state.
- (b) Write out $|+\rangle|+\rangle$, which is what a successful copy of $|+\rangle$ would be. Compare the two expressions and say what you conclude about C .
- (c) ★ Your argument used a particular pair of states that C copies. For which pairs of states $|a\rangle, |b\rangle$ is a copier possible? Justify your answer, and relate it to why error correction cannot proceed by storing duplicates.

6. ★ Platforms. (facultative)

- (a) State DiVincenzo's criteria.
- (b) Pick any two of the platforms mentioned in the lecture. For each, name the criterion it satisfies most comfortably and the one that causes it the most difficulty, in one line each.
- (c) A two-qubit gate that fails once in 10^3 attempts is among the best available, while a long algorithm needs failure rates many orders of magnitude smaller. Say briefly what closes that gap, and what the word *threshold* refers to.

Challenge

7. Turning polarisation by measuring it. Two polarising filters crossed at 90° transmit no light, and inserting a third at 45° between them lets some through. Generalise the arrangement: place N filters between the crossed pair and the source, each rotated by a further $90^\circ/N$ from the one before it, so that the last filter is at 90° to the first. A photon polarised along the axis of one filter passes the next, which is rotated by ϑ , with probability $\cos^2 \vartheta$.

- (a) Justify the claim that the transmission of the whole stack, for light already polarised along the first filter, is $T(N) = \cos^{2N}(\pi/2N)$.
- (b) Evaluate T for $N = 2$, $N = 10$ and $N = 100$. Check that $N = 2$ reproduces the familiar three-filter result.
- (c) Expand $\cos(\pi/2N)$ for large N and show that $T(N) \approx \exp(-\pi^2/4N)$. Hence find $\lim_{N \rightarrow \infty} T(N)$.
- (d) The polarisation has been rotated through a right angle, and in the limit no light is lost. Reconcile this with the fact that each filter performs a measurement, and that a measurement was supposed to destroy information about the previous one.